

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

3 Computation Theory (amp12)

- Register machines (a) Define what is meant by a *configuration* and a *computation* of a register machine (RM). Explain carefully what it means to say that a computation *halts*.

[5 marks]

Answer: A *configuration* $(\ell, [r_0, r_1, \dots, r_n])$ of a register machine consists of a number $\ell \in \mathbb{N}$ (position of the current program instruction in the program) together with a list of numbers $[r_0, r_1, \dots, r_n]$ (current contents of all the machine's registers $R_0, R_1, \dots, R_n$).

A *computation* of a RM is a (finite or infinite) sequence $c_0, c_1, c_2, \dots$ of configurations such that c_0 is *initial*, meaning that $c_0 = (\ell, [r_0, r_1, \dots, r_n])$ with $\ell = 0$, and each subsequent element c_{i+1} of the sequence is obtained from the preceding element $c_i = (\ell, [r_0, r_1, \dots, r_n])$ by carrying out the RM's program instruction labelled L_ℓ with registers $R_0, R_1, \dots, R_n$ containing $r_0, \dots, r_n$.

A computation *halts* if it is a finite sequence $c_0, \dots, c_n$ and the last configuration $c_n = (\ell, [r_0, r_1, \dots, r_n])$ is a *halting configuration*, meaning that either there is a HALT instruction labelled L_ℓ in the RM's program, or ℓ is greater than the length of the list of instructions in the RM's program (so that there is no instruction labelled L_ℓ).

- (b) Define the notion of RM *computable* partial numerical function of n arguments.

[2 marks]

Answer: $f \in \mathbb{N}^n \rightarrow \mathbb{N}$ is register machine computable if there is a register machine M with at least $n + 1$ registers $R_0, R_1, \dots, R_n$ (and maybe more) such that for all $(x_1, \dots, x_n) \in \mathbb{N}^n$ and all $y \in \mathbb{N}$, the computation of M starting with $R_0 = 0, R_1 = x_1, \dots, R_n = x_n$ and all other registers set to 0, halts with $R_0 = y$ if and only if $f(x_1, \dots, x_n) = y$.

- Undecidability of the halting problem (c) What does it mean for a problem (expressed as a property of numbers) to be RM *undecidable*?

[2 marks]

Answer: $S \subseteq \mathbb{N}$ is register machine decidable if its characteristic function $\chi_S : \mathbb{N} \rightarrow \mathbb{N}$ $\left(\chi_S(n) \triangleq \begin{cases} 1 & \text{if } n \in S \\ 0 & \text{if } n \notin S \end{cases} \right)$ is register machine computable.

- (d) A computation of a RM is said to be *circular* if it reaches the same configuration at two different times.

- (i) Explain why a circular computation does not halt. Give an example of a RM computation that does not halt, but that is not circular. [2 marks]

Answer: If a RM computation $c_0, c_1, c_2, \dots$ has the property that $c_i = c_j$ with $i < j$, then since RM computation is deterministic (the next configuration in a computation is uniquely determined by its predecessor in the sequence, if the latter is non-halting), the computation must look like $c_0, \dots, c_i, \dots, c_j = c_i, \dots, c_j = c_i, \dots$ and is infinite. So circular computations are not halting computations.

Consider the RM whose program has a single instruction, labelled L_0 , that increments R_0 and then jumps to L_0 . It only has computations of the form $(0, [r_0]), (0, [r_0 + 1]), (0, [r_0 +$

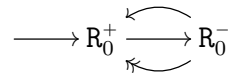
2]), ... that are infinite (so non-halting), but all of whose configurations are distinct (so non-circular).

- (ii) The Circularity Problem is: Decide whether or not the computation of any given RM and initial register contents is circular. Give a proof that the Circularity Problem is undecidable. You may assume suitable functions for encoding and decoding pairs of numbers as numbers, finite lists of numbers as numbers, and RM programs as numbers. [9 marks]

Answer: Some notation: let $prog(e)$ denote the RM program with code $e \in \mathbb{N}$, let $\ulcorner [x_0, \dots, x_n] \urcorner \in \mathbb{N}$ denote the code of a list of numbers $[x_0, \dots, x_n]$, and let $\langle x, y \rangle \in \mathbb{N}$ denote the code of a pair of numbers (x, y) . Let $C(e, a)$ stand for the statement “ a decodes as the list $[a_0, \dots, a_n]$ and the computation of the RM program $prog(e)$ started with $R_i = a_i$ for $i = 0, \dots, n$, and all other registers set to 0, is circular”. Let $H(e, a)$ stand for the statement “ a decodes as the list $[a_0, \dots, a_n]$ and the computation of the RM program $prog(e)$ started with $R_i = a_i$ for $i = 0, \dots, n$, and all other registers set to 0, halts”

We are required to show that $S \triangleq \{\langle e, a \rangle \mid C(e, a)\}$ is a RM undecidable subset of $\mathbb{N}$. Here are two possible approaches (there may be more).

- (A) *Reduce to the known undecidability of the halting problem.* Let $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be the the RM computable total function given by the RM that takes a single argument e , changes the program $prog(e)$ by replacing each *HALT* instruction (and each jump to a label with no instruction) with the program



(which is always circular), and then encodes that program to obtain the value of φ at e .

Note that for all $e, a \in \mathbb{N}$ we have

$$\begin{aligned} C(e, a) &\Rightarrow \neg H(e, a) \\ H(e, a) &\Rightarrow C(\varphi(e), a) \\ C(\varphi(e), a) &\Rightarrow (H(e, a) \vee C(e, a)) \end{aligned}$$

and hence

$$\begin{aligned} (C(e, a) \wedge C(\varphi(e), a) &\Rightarrow \neg H(e, a) \\ (C(e, a) \wedge \neg C(\varphi(e), a) &\Rightarrow \neg H(e, a) \\ (\neg C(e, a) \wedge C(\varphi(e), a) &\Rightarrow H(e, a) \\ (\neg C(e, a) \wedge \neg C(\varphi(e), a) &\Rightarrow \neg H(e, a) \end{aligned}$$

If we had a RM to decide membership of S , since φ is a RM computable total function, by composition we would also have a RM to decide membership of $\{\langle e, a \rangle \mid C(\varphi(e), a)\}$. Using these machines to computer whether or not $C(e, a)$ holds and whether or not $C(\varphi(e), a)$ holds, the above four implications show that

in each of the four cases we can then decide whether or not $H(e, a)$ holds. But it is a standard result that $\{\langle e, a \rangle \mid H(e, a)\}$ is RM undecidable. So there can be no RM to decide membership of S .

- (B) *Adapt the proof of undecidability of the halting problem.* We suppose M is a register machine that decides membership of S and derive a contradiction. Let M' be the machine obtained from M by replacing each *HALT* instruction (and each jump to a label with no instruction) with the program

$$\longrightarrow R_0^- \longrightarrow \text{HALT}$$

which is circular if $R_0 = 0$ and halts if $R_0 = 1$.

Let M'' be the machine obtained from M' by adjoining instructions at the beginning of its program that replace the contents x of R_1 with the coded pair $\langle x, \ulcorner x \urcorner \rangle$. Finally, let e be the code of the program of M'' .

For any $x \in \mathbb{N}$, by assumption that M decides membership of S we have: M started with $\mathbf{R}_1 = \langle e, x \rangle$ and all other registers set to 0 always halts, with $\mathbf{R}_0 = 1$ if $\langle e, x \rangle \in S$ and $\mathbf{R}_0 = 0$ if $\langle e, x \rangle \notin S$. Hence by construction of M' from M we have that M' is circular when started with $\mathbf{R}_1 = \langle e, x \rangle$ and all other registers set to 0 iff $\langle e, x \rangle \notin S$. Therefore, setting $x = e$, by construction of M'' from M' we have:

- (I) M'' is circular when started with $R_1 = e$ and all other registers set to 0 iff $\langle e, \lceil [e] \rceil \rangle \notin S$.

Note that by definition of S , $\langle e, \lceil [e]^\lceil \rangle \in S$ iff *prog*(e) is circular when started with $R_1 = e$ and all other registers set to 0. So by definition of e we also have:

- (II) $\langle e, \ulcorner [e] \urcorner \rangle \in S$ iff M'' is circular when started with $\mathbf{R}_1 = e$ and all other registers set to 0.

From (I) and (II) we have $\langle e, \ulcorner [e] \urcorner \rangle \notin S$ iff $\langle e, \ulcorner [e] \urcorner \rangle \in S$, which is a contradiction. Therefore no such RM M can exist and S is RM undecidable.